WHAT IS INPUT LAG?

In the simplest of terms, it's the bane of gamers everywhere. A parameter that renders perfectly good monitors absolutely unusable for gaming. Input lag is essentially the delay caused by monitor's own video processor between receiving the signal, processing it, and then outputting it to the panel. Gamers, on the other hand, measure input lag in terms of the delay between pressing a button and the same being translated onto the screen.

- **Faster response time** – Due to its reduced complexity compared to other display technologies like IPS and VA, the TN displays can change state very quickly and thus are highly preferred by gamers as response time is paramount in gaming, especially online gaming. The introduction of Response Time Compensation (overdrive) drove down the response time even lower and now some TN displays can support true 120+ Hz refresh rates.

Cons

- **Comparatively poor colour reproduction** – As mentioned earlier, TN panels use 6-bits per RGB colour, that is 18 bits in total and are unable to display 24-bit true colour that is received from graphics cards. Instead, TN displays use a 24-bit dithering process that which can be either special (combines adjacent pixels to produce the desired colour shade) or temporal (cycles between different shades to simulate intermediate shade). Both tend to make the individual pixels visible.
- **Narrow Viewing angles** - Compared to more expensive displays, TN panels fall short in the department of viewing angles. It doesn't take much effort to see the screen discolour. It just has to be viewed from a sufficiently steep angle. For example, viewing the picture from below can make the image appear quite dark and the opposite effect is achieved if you try to view the display from the top.